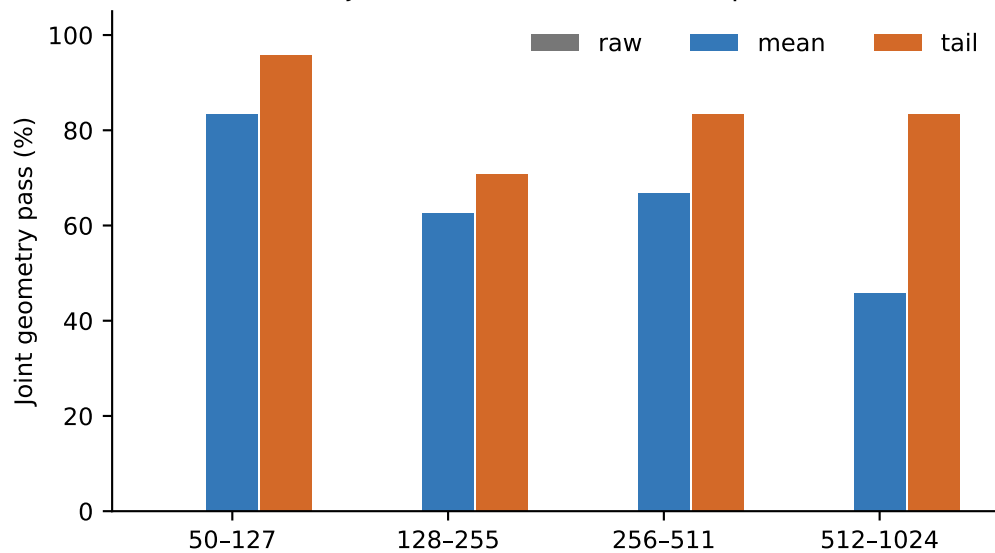
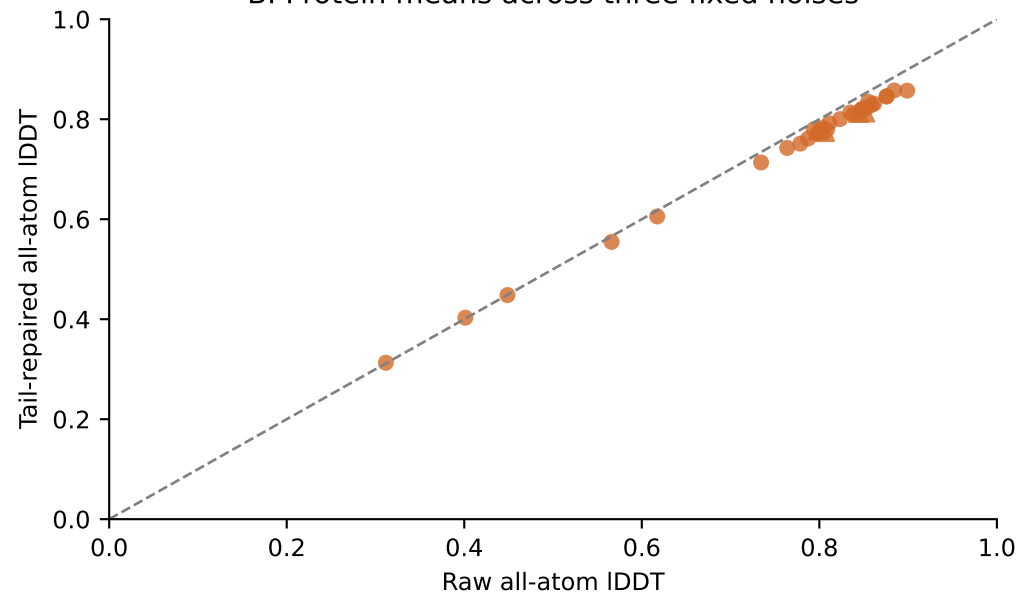


C4/S1 + frozen iterative geometry repair: independent32

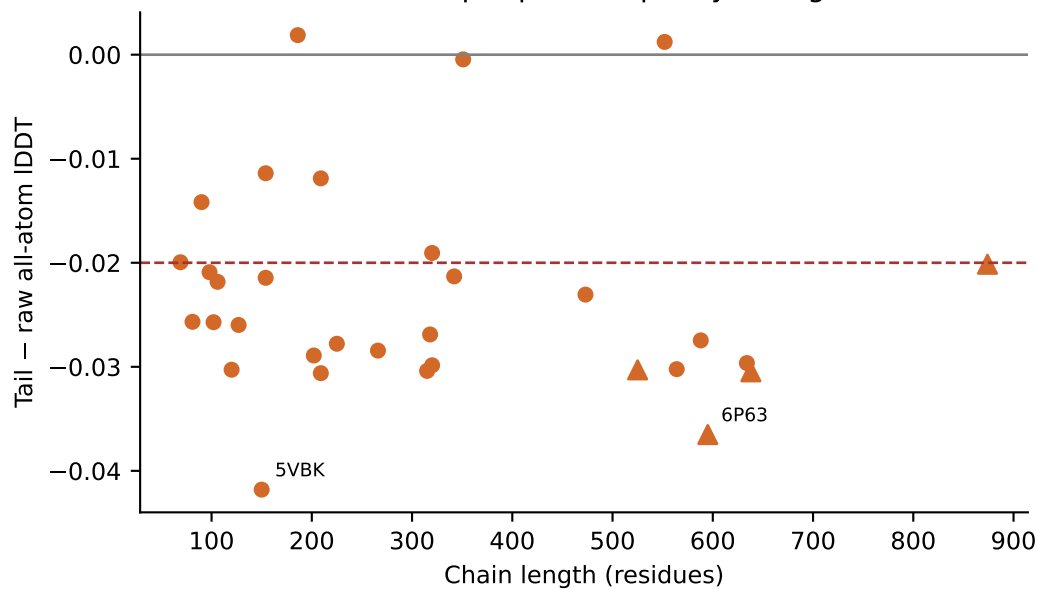
A. Locked joint checks (24 instances per stratum)



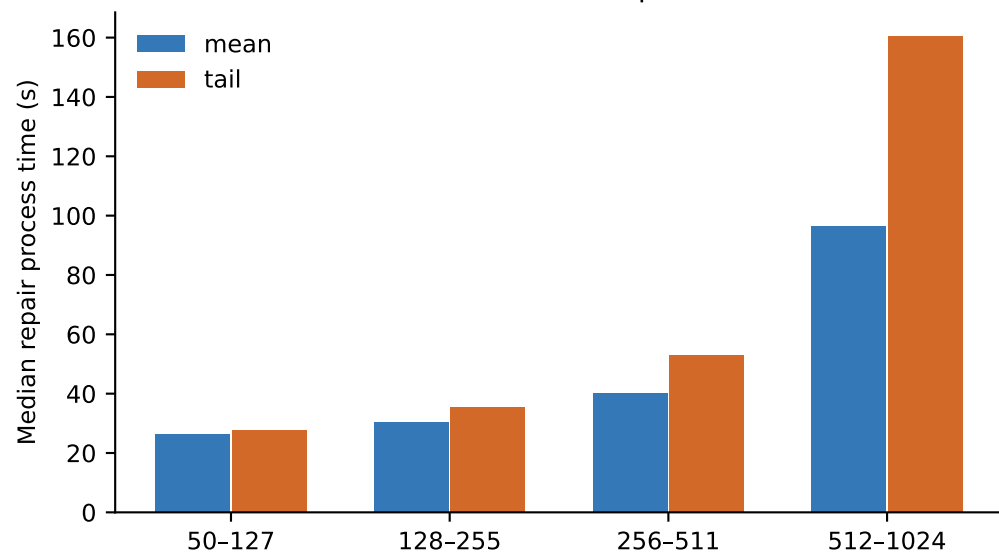
B. Protein means across three fixed noises



C. Paired per-protein quality change



D. Actual cost; iteration caps are matched



Triangles: four homooligomer-derived chains. Same 28+4 source panel and all fixed seeds; no best-of-three.
Operational geometry checks do not imply complete chemical validity, one-step repair, or design utility.